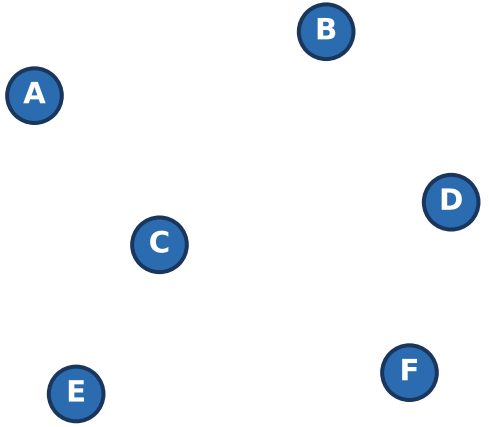
Paradigm Shift: From Metabolites to Reactions

Reactomics treats chemical reactions as first-class analytes in untargeted MS data

Traditional Metabolomics



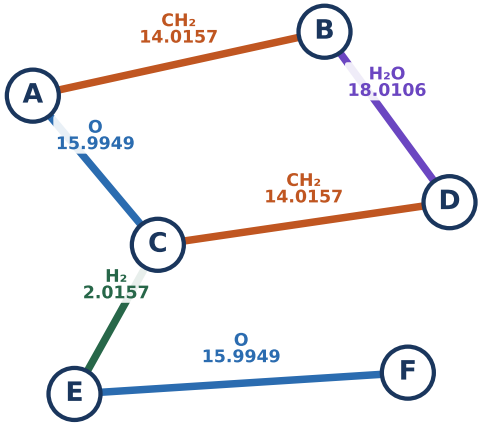
Metabolites as independent entities

Question: "Which metabolites differ?"

Requires compound annotation

SHIFT

Reactomics



Reactions (PMDs) as first-class objects

Question: "Which reactions differ?"

No compound ID required

PMD-based reactomics · Communications Chemistry 2020 · pmd R package